Forces and free motion

Measure $x(t)$ from the equilibrium position.

Spring force

$$F_s = -kx$$

Damping force

$$F_d = -bx'$$

Newton's second law connects both forces.

$$mx'' = -bx' - kx$$

$$mx'' + bx' + kx = 0$$

$m > 0$, $k > 0$, $b > 0$; constant coefficients; no driving force.

Next: which function satisfies inertia, damping and restoring force?

Why oscillations decay

Try an exponential to turn differentiation into algebra.

$$x(t) = Ce^{\lambda t}$$

$$m\lambda^2 + b\lambda + k = 0$$

$$\gamma = \frac{b}{2m}, \qquad \omega_0 = \sqrt{\frac{k}{m}}$$

Underdamped condition: $0 < \gamma < \omega_0$.

$$\lambda = -\gamma \pm i\omega_\mathrm{d}, \qquad \omega_\mathrm{d} = \sqrt{\omega_0^2 - \gamma^2}$$

Real part: decay. Imaginary part: oscillation.

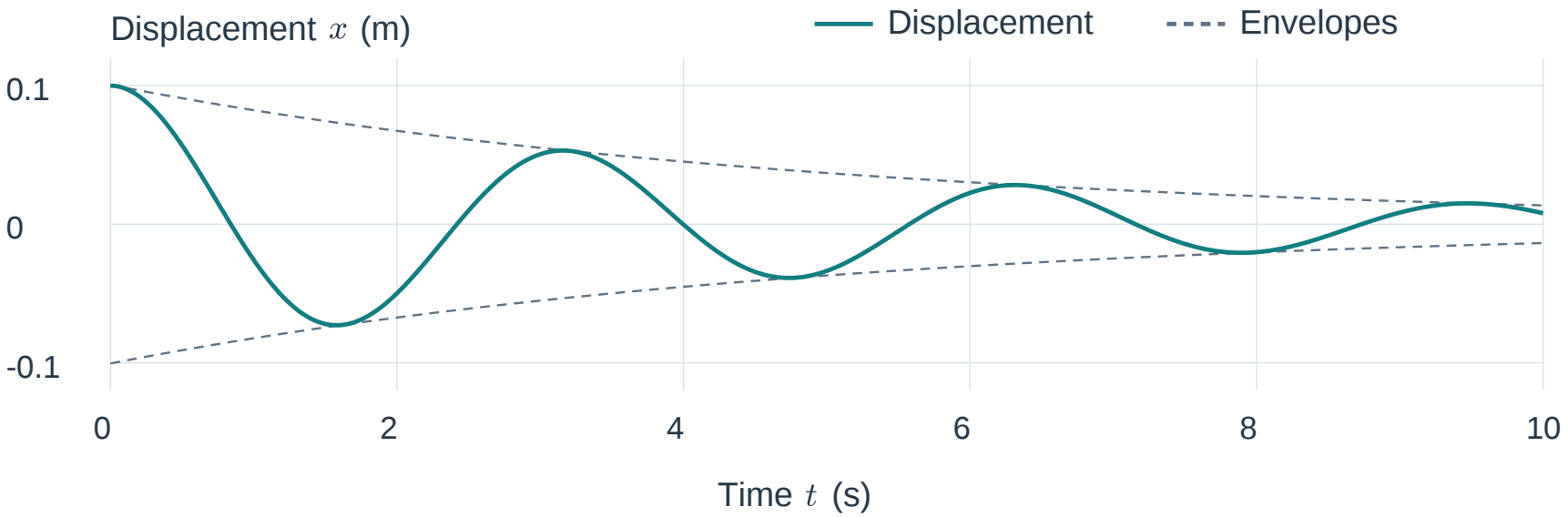
Next: how do initial displacement and velocity select the motion?

Initial data choose the motion

$$x(t) = e^{-\gamma t}\left[A\cos(\omega_{\mathrm{d}} t) + B\sin(\omega_{\mathrm{d}} t)\right]$$

$$A = x_0, \qquad B = \frac{v_0 + \gamma x_0}{\omega_{\mathrm{d}}}$$

Example: $m = 1$ kg, $b = 0.4$ kg/s, $k = 4$ N/m, $x_0 = 0.1$ m, $v_0 = 0$.



Envelopes: $\pm R e^{-\gamma t}$, $R = \sqrt{A^2 + B^2}$. Displacement still oscillates.